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Word Segmentation using Self-Supervised Hierarchical Transformers for *Scriptio Continua* in Greek and Latin

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Abstract

The digitization of historical documents written in *Scriptio Continua*, a style of writing without spaces between words, requires not only accurate Optical Character Recognition (OCR), but also a second step of very costly manual segmentation to convert character sequences into meaningful words. This article presents a new self-supervised hierarchical transformer model that automates word segmentation for *Scriptio Continua* in Greek and Latin texts. Our method utilizes readily available printed editions to train, first, a character-level transformer for word boundary labeling and, second, a word-level transformer to reclassify segmentation candidates based on a perplexity score. Evaluated on a diverse corpus of Greek and Latin codices (4th-6th centuries) and Byzantine seals (6th-12th centuries), our approach significantly outperforms state-of-the-art Bayesian (NHPYLM) and flat neural network models, achieving F1 scores of 96%, 94%, and 42% for boundary detection, respectively, on the three considered datasets. While significantly improving the performance of current approaches on highly abbreviated texts, this study highlights ongoing challenges for the study of abbreviations on smaller epigraphic objects, such as seals and coins. We additionally provide our code and datasets to ensure full reproducibility and foster future research.

1 Introduction

From Antiquity to the early medieval times, texts were copied in *Scriptio* (or *Scriptura*) *Continua*, a writing style without spaces between words. The text was laid out continuously on a line without word division, and was written solely using the majuscule script (or *uncial* script), and missed punctuation and diacritics (Cavallo (2008)). Even when the early Middle Ages began to develop the *minuscule* script that eventually replaced the uncial script, *Scriptio Continua* continued to be used for inscriptions on objects where space was limited and had to be optimized, such as the seals

and coins that circulated in the Byzantine Empire (Nesbitt (2008)).

Recent years have seen growing performance in the quality of *Optical Character Recognition* (OCR) and *Handwritten Text Recognition* (HTR) approaches for the automatic retrieval of textual content written, inscribed, or carved on ancient materials (Torres Aguilar and Jolivet (2023); Vidal-Gorène et al. (2025)). However, this output can often remain noisy, especially in the case of historical documents, and post-OCR correction relying on recent advances in *Natural Language Processing* (NLP) has become an active field of research (Thomas et al. (2024); Boros et al. (2024);

Yasin et al. (2023)).

For artifacts written in *Scriptio Continua*, the challenge of automatic transcription extends beyond character-recognition accuracy. The absence of word boundaries in the source material itself prevents the transition from character recognition to lexical units, making *word segmentation* a necessary and non-trivial step in transforming raw OCR output into a readable text. Thus, even in the ideal scenario of perfect character transcriptions, segmenting these sequences requires heavy, extensive manual intervention, and this segmentation step remains a bottleneck for the digitization of ancient artifacts. While this difficulty has been acknowledged as one of the most common challenges when transcribing Ancient Codices (Perdiki (2023); Geldhauser and Malyshev (2024); Kolak et al. (2003); Nguyen et al. (2021)), post-OCR *word segmentation* of the transcriptions is rarely mentioned as a necessary step in the OCR pipeline and is treated as a downstream task. A notable exception is the work of Clérice (2020), which employs a fully supervised LSTM architecture for segmenting Latin and Old French manuscripts, a task that requires extensive labeled data.

This paper addresses the word segmentation stage for texts written in *Scriptio Continua*, considered the final step in the OCR pipeline. We introduce a new self-supervised transformer-based algorithm for segmenting the post-OCR output of *Scriptio Continua* transcriptions. Unlike the fully supervised existing approaches, our method is trained on available printed editions and requires no manually segmented training data, making it particularly suitable for domains where manually tagged data is scarce. To isolate the segmentation problem itself, we operate under the assumption of accurate character-level OCR output and evaluate only the word segmentation performance, regardless of the OCR quality. This framing enables us to focus solely on the possibility of segmenting continuous character sequences of Greek and Latin into words, rather than enhancing the OCR quality itself.

We evaluate the proposed approach on three representative artifacts from Late Antiquity to the Early medieval period: Greek and Latin Codices and Byzantine Seals, and demonstrate that our approach outperforms the current baselines for word segmentation without requiring annotated

training data. Figure 1 presents three representative examples, one of each kind of artifact, showing the input data, the expected output, and the corresponding English translation for each transcription.

The main contributions of this study are:

- A new self-supervised hierarchical method for segmenting *Scriptio Continua* Greek and Latin texts, that achieves an F_1 score for word boundary detection of 96% (4 Greek codices), 94% (7 Latin Codices), and 42% (699 inscriptions on Byzantine Seals).
- A comprehensive benchmark comparing Bayesian (NHPYLM) and character labeling approaches to our proposed solution, across these three datasets. Compared to standard character classification models, our approach yields F_1 gains of 2.1% for word boundary detection and 41.2% for NHPYLM (Greek codices). For Latin codices, it achieves gains of 3.3% and 32.4%, and for Byzantine seals, it achieves gains of 42.9% and 31.3%.
- The open-source release of our algorithm’s implementation and the corresponding benchmarking datasets to ensure reproducibility and encourage further work.

The paper is organized as follows. Section 2 contains the assumptions made in the study and the mathematical formalization of the problem. Section 3 presents the current state of the art regarding existing approaches for word segmentation, and Section 4 presents our proposed method. Section 5 details the data used in the study, as well as the self-supervised framework we designed to train our models. Section 7 presents the performance of our algorithm, along with a comparison to the current state of the art. We provide a qualitative analysis of the errors made by the algorithm in Section 8 and summarize the limitations of our approach in Section 9. Section 10 concludes this paper and provides our perspectives for further research work.

2 Problem framing and formalization

2.1 Study assumptions

We base our work on the following hypotheses, which we take for granted throughout this study:

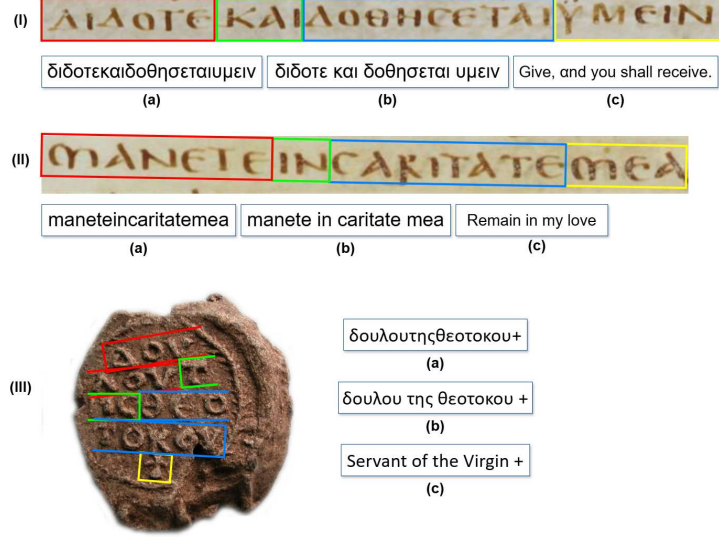


Figure 1: Three examples of Scriptio Continua writings, taken from (i) The *Bezae* Codex, Luke 8:38 in Greek, (ii) The *Bezae* Codex, John 15:10 in Latin, and (iii) Byzantine seal of Theodoros *apo eparchon* from the Tatiş collection (Cheynet and Tatiş 2019, p. 196). Each example illustrates (a) the transcription provided as input, (b) the expected output produced by the model, and (c) the corresponding translation for reference.

1. **Assumption 1:** The input to our model is the output of a prior OCR process. We assume this output is a character-accurate transcription of the source text, containing no recognition errors. This problem framing is a deliberate choice to isolate the already challenging word segmentation task and evaluate the performance of our algorithm on a perfect dataset as a proof-of-concept study, before investigating the complexity of the noise generated by OCR.
2. **Assumption 2:** The OCR transcriptions are provided as plain character sequences and do not contain any layout or typographical information from the considered artifact.
3. **Assumption 3:** Sentence segmentation is considered beyond the scope of this study, since this step reflects modern editorial conventions and is absent from the historical artifacts under analysis.

2.2 Problem formalization

We define the word segmentation problem as follows.

Let Σ be an alphabet of characters. For an input string $S = c_1 c_2 \dots c_n$, a *segmentation* is defined by

a strictly increasing sequence of boundary indices

$$B = \langle b_0, b_1, \dots, b_k \rangle, \quad 0 = b_0 < b_1 < \dots < b_k = n,$$

which partitions S into k contiguous substrings

$$w_j = c_{b_{j-1}+1} \dots c_{b_j}, \quad j = 1, \dots, k.$$

In our framework, we consider w_j to be the *segmented words*, and $S = w_1 w_2 \dots w_k$.

If for each $n \geq 0$,

$$\mathcal{B}_n = \{ \langle b_0, b_1, \dots, b_k \rangle \mid 0 = b_0 < b_1 < \dots < b_k = n \}$$

is the set of valid boundary sequences for strings of length n , then the *segmentation function* is a mapping

$$f : \Sigma^* \rightarrow \bigcup_{n \geq 0} \mathcal{B}_n$$

which assigns to each string S of length n a valid boundary sequence $f(S) \in \mathcal{B}_n$.

The goal of this paper is thus to propose a *segmentation function* f that, for a given string S of size n , selects the most probable valid boundary sequence $B \in \mathcal{B}_n$, given the string S , i.e., $B^* = \arg \max_{B \in \mathcal{B}_n} \mathbb{P}(B \mid S)$.

3 Word segmentation in the state of the art

Unsupervised Bayesian approaches for identifying the most probable segmentation from a language model have historically been among the most extensively researched methods for word segmentation. Several models for language modeling have been proposed, ranging from hierarchical or non-hierarchical Dirichlet processes (Goldwater et al. (2006)) to a hierarchical Pitman-Yor model (Teh (2006)). In this study, we focus on the Nested Hierarchical Pitman-Yor Language Model (NHPYLM) (Mochihashi et al. (2009)) as a baseline for Bayesian models for word segmentation. This approach operates at both character and word levels, thereby facilitating the learning of the lexicon and the probabilities of its sequences. The first-level models character sequences to assign a probability to each segment being a word; the second-level models word sequences, using the segments that the first layer makes probable as a lexicon. The optimal segmentation then corresponds to the sentence segmentation that maximizes the posterior probability, allowing a balance between lexical coherence (favored by the character model) and the regularity of word sequences (captured by the model at the lexical level). Consequently, the most probable segmentation is that which maximizes the posterior likelihood of the words in their given order, given the inferred lexicon.

However, although these models have demonstrated effectiveness across several benchmarks, specifically Chinese segmentation (Xu et al. (2008)), Japanese (Mochihashi et al. (2009)), and very recently Tamil (Sandeep et al. (2025)), they have several limitations: **(1)** they have limited context windows since they are *n-gram* models, and cannot capture dependencies and ambiguities that are resolved over long sentences, a feat achievable by neural networks; **(2)** they do not integrate semantic concepts into their language modeling: each word is a distinct symbol and is not mapped into a continuous semantic space as neural networks are capable of doing.

Recent advances in neural NLP have framed word segmentation as a sequence labeling issue (Xue (2003)), where boundaries are the labels to predict. For example, Shao et al. (2018) proposes using this sequence labeling approach with the

BiRNN-CRF model, which operates at the character level and trains its model on the Treebank. While these methods have been used with success on several languages that do not naturally provide word boundaries, such as Javanese (Nevin et al. (2024)) and Chinese (Chen et al. (2015)), and even Greek and Latin (Shao et al. (2018)), they present significant limitations for our study. First of all, with the exception of Clérice (2020), their application has been limited to modern languages or standardized printed editions of ancient texts, which lack the complex morpho-syntactic variations, abbreviations, and deterioration found in ancient artifacts. Additionally, their architecture is limited to character-level sequence labeling, which fails to leverage word-level cues, unlike the NHPYLM approach. This inherently leads to a lack of accounting for the semantic coherence of the returned word sequence. Finally, the studies performed on these models, operating within a supervised framework, relied on an annotated subset of the considered texts, whether in printed editions or manuscripts.

This paper bridges this gap by utilizing a novel hierarchical neural architecture based on the transformer (Vaswani et al. (2017)). Our model relies on two components: a character-level transformer that encodes each character and predicts potential word boundaries, and a word-level transformer that scores the likelihood of the sequence of segments. This enables us to leverage both character patterns and word-level context, as NHPYLM does. These transformers are then trained in a self-supervised manner using segmented printed editions before being used for the segmentation of artifacts that the model has never seen during the training phases.

4 Hierarchical transformers for word segmentation

In this study, we thus propose an original hierarchical neural architecture that adapts the intuition of Bayesian hierarchical models while leveraging the representational power of transformers.

Our approach utilizes two probabilistic neural components: a character-level model that operates at the character level to generate potential word segmentations through sequence classification for

word boundary detection, and a word-level model that evaluates sequence coherence.

4.1 General workflow

Our segmentation pipeline consists of a two-step inference process. The first stage follows the approach of the Universal Segmenter: we train a neural model (the *character model*) that predicts, for each character, whether or not it is followed by a space, which then allows us to retrieve its boundaries. The embedding is performed directly at the character level, and the model outputs a label l_i for each character c_i . We test two types of labeling (binary or multi-class) to represent the labels of the word boundaries to be predicted, as detailed in Section 5.2.

However, unlike the Universal Segmenter, we enhance this approach by employing a beam search during inference. Instead of selecting only the top-scoring segmentation, we output the k most probable word segmentation according to the model. This initial prediction stage, therefore, allows us to generate k segmentation hypotheses. The second stage introduces a novel re-ranking step for the k hypotheses of the previous step, utilizing an autoregressive *word model*. This model, a transformer trained on the same corpus as the character level, operates directly at the word level. It evaluates the linguistic plausability of the k candidate segmentations, which we measure by calculating the sequence probability (or, equivalently, the perplexity) of each segmentation. The final segmentation is selected among the k candidates exclusively based on the model’s perplexity score at the word level, without considering the scores of each beam from the previous step. This two-step process effectively separates local evidence from global evidence of lexical coherence. A concrete example of its application is shown in Figure 2.

4.2 Segmentation at the character level

Prediction of character tags

We consider the character-level segmentation task as a sequence labeling problem, using a standard transformer encoder architecture to implement the sequence classification.

Given the character sequence $S = c_1c_2\dots c_n$ to be segmented, each character c_t is individually projected into a vector space of dimension d . We thus perform the embedding directly at the character level, rather than at the word/token level as is typically done. After the processing of the transformer layers, we obtain a contextualized representation for each character c_t :

$$\mathbf{h}_t \in \mathbb{R}^d$$

The final classification layer consists of a linear transformation followed by a softmax function, yielding $\mathbf{p}_t \in \mathbb{R}^Q$, the probability distribution over the Q possible labels for the classification for the character c_t , as specified in Section 5.2.

The prediction for each character is then the most likely class based on the probability distribution for the last layer:

$$\hat{s}_t = \arg \max_{k \in [1, Q]} \mathbf{p}_t[k]$$

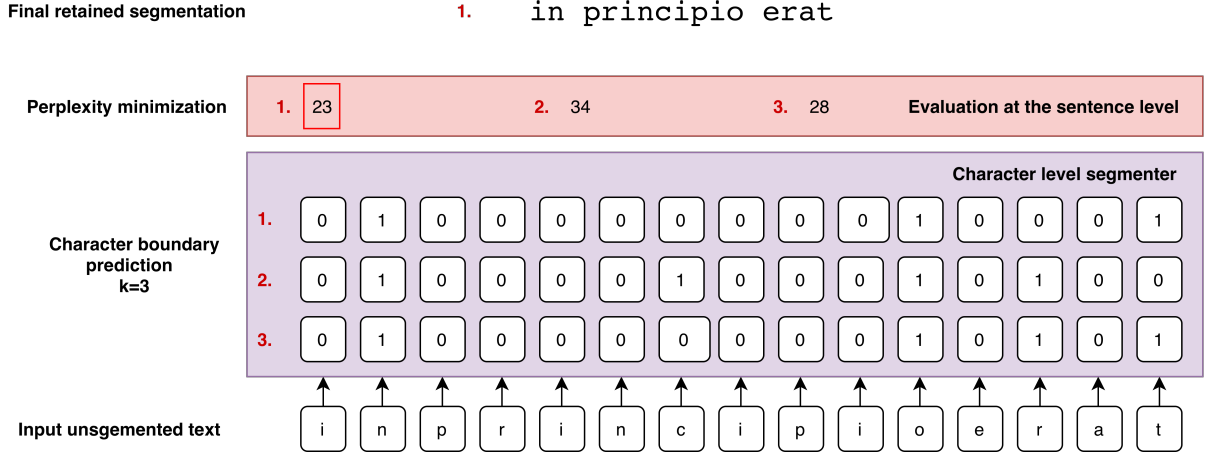
The model is trained end-to-end by minimizing the cross-entropy loss between the predicted distributions and the target labels across the whole sequence.

Beam search

Instead of proposing a single segmentation for a given sentence S , *i.e.*, the most probable one, we opt for a beam search approach. This search heuristic enables us to propose the k most probable segmentations (with k referred to as the beam width) for the classification model under consideration, while maintaining several hypotheses in parallel throughout the classification procedure. In this paper, we adopt the original procedure (Freitag and Al-Onaizan (2017)).

This process consists of maintaining, for each prediction made character by character, the k best predicted sequences, known as beams. At the end of the procedure, when all characters have been processed, the k best solutions are returned. Thus, we calculate the score for a candidate label sequence $L_{1:t} = (l_1, \dots, l_t)$ as the joint log-likelihood of the sequence, with the probability of each label provided by the character-level model conditioned

Figure 2: Example of application of the proposed pipeline for segmenting the string "inprincipioerat"



on the input string S .

$$\begin{aligned} \mathcal{L}(L_{1:t}) &= \log \mathbb{P}(L_{1:t} | S) \\ &= \sum_{i=1}^t \log \mathbb{P}(l_i | S). \end{aligned}$$

where $\mathbb{P}(l_i | S)$ is the probability distribution over labels for the i -th character output by the character model discussed previously.

In the next step $t + 1$, we extend each candidate sequence with each of the possible labels for l_{t+1} , and we update the score of the new sequence of labels:

$$\mathcal{L}(L_{1:t+1}) = \mathcal{L}(L_{1:t}) + \log \mathbb{P}(l_{t+1} | S)$$

Among all the extended sequences, only the k with the highest scores are kept for the next step. After scanning the entire string, we obtain the k most probable segmentation according to the beam-search heuristic. k is one of the hyperparameters of our model, and we will evaluate its impact in Section 7.2.

4.2.1 Evaluating segmentation at the word level

The second step in our workflow utilizes a pre-trained autoregressive language model (Liu et al. (2022)) not for generation, but as a scoring mechanism to evaluate the linguistic plausibility of the

k candidate segmentations produced by the previous step.

Given the i -th of the k candidate segmentations $\mathcal{B}_i$, the autoregressive model, pre-trained on the corpus in question, allows us to score the likelihood of the sequence according to the linguistic distribution it learned on the training corpus. To quantify this likelihood, we use the *perplexity* metric (PPL), a standard measure for measuring the uncertainty of a language model (Jelinek et al. (2005)). This metric is defined as the exponentiated average negative log-likelihood of a token sequence, the candidate segmentation itself having been tokenized. Thus, if $x_1, x_2, \dots, x_m$ is the sequence of tokens resulting from the segmentation B , the perplexity is given by:

$$\begin{aligned} \text{PPL}(B) &= \exp(\mathcal{L}(B)) \\ &= \exp\left(-\frac{1}{m} \sum_{t=1}^m \log \mathbb{P}(x_t | x_{1:t-1})\right) \end{aligned}$$

where $\mathcal{L}(B)$ is the average negative log-likelihood returned by the model for the sequence of words $w_1 w_2 \dots w_n$ generated by the segmentation B , with $\mathbb{P}(x_t | x_{1:t-1})$ is the probability assigned by the model to token x_t given all previous tokens $x_{1:t-1}$, and m is the total number of tokens.

A lower perplexity indicates that the segmented sequence B is more probable and more consistent with the language model, thus serving as a confidence measure for the segmentation. We therefore

take, among the k possible segmentations, the segmentation $\hat{B}$ that minimizes the perplexity of the autoregressive model:

$$\hat{B} = \arg \min_{B \in \mathcal{B}} \text{PPL}(B)$$

where $\mathcal{B}$ denotes the set of k candidate segmentations. $\hat{S}$ is then the segmentation returned by our segmentation function f .

5 Self-supervised framework

5.1 Learning from printed text

While NHPYLM is entirely unsupervised, our approach, which relies on classification, is supervised and therefore requires training data. However, when working with ancient artifacts that have been little studied, this data is often unavailable, hard to access, and expensive to annotate. This may therefore lead to the risk that supervised methods are less effective than unsupervised approaches. To address this data availability challenge, we employ a self-supervised training strategy. Rather than using manual annotations, we generate training labels automatically by utilizing the spaces from available digital editions. As a result, this study does not require any human intervention for annotating the ground truth used, making our approach highly scalable.

However, this choice of training data introduces a new challenge, as the linguistic and orthographic characteristics of the printed editions differ from those present in the actual manuscripts. To evaluate the resilience of our method to this distribution shift between training and test data, we use three distinct use cases:

1. Training on an eclectic critical edition of the Greek New Testament (dataset **CodGreek**);
2. Training on a text standardized after the manuscripts considered for the Latin New Testament (dataset **CodLatin**);
3. Training on transcriptions from a catalog for a set of Byzantine seals (dataset **Seals**)

For the codices, segmentation is done at the verse level, and for the seals at the transcription level. This diversity between the training and test sets allows us to evaluate our model’s capacity to generalize across different text sources. A summary

of the characteristics of the datasets considered is presented in Table 1.

Training on a critical edition (CodGreek)

Our first use case is the Gospel of Luke, a book from the New Testament, in its Greek version. This text, and more generally the Gospels, are among the most copied texts in Western history in their Greek and Latin versions, and are therefore among the most relevant manuscripts to be processed by automated approaches, due to the enormous volume of available data. In this study, we consider four codices, copied in *Scriptio Continua* between the 4th and 6th centuries AD, as test data. This provides a test set of 4,600 verses from Greek codices for segmentation.

For this use case, the training data does not come from the manuscripts themselves, but from an eclectic, copyright-free edition called the SBLGNT. This edition contains a ”probable” original text reconstructed by philologists, and therefore does not correspond to a given manuscript, but borrows variants from all the manuscripts in the study (Maas (1958)).

An example of a distribution shift between the train and test data can be seen in the first part of verse 8, chapter 5 of the Gospel of Luke in the SBLGNT, and in two of the four codices in our studies, where we see a text that varies:

- **Printed text: SBLGNT¹** :
 ἰδὼν δὲ Σίμων Πέτρος προσέπεσεν τοῖς γόνασιν Ἰησοῦ [Then seeing, Peter fell at the knees of Jesus]
- **Real manuscript 1: Vaticanus** :
 ἰδὼν δὲ σίμων πετρος προσεπεσε τοις γονασιν ιω [Then seeing, Peter fell at the knees of Jesus]
- **Real manuscript 2: Bezae** : ο δε σιμων προσεπεσεν αυτου τοις ποσιν [Then Simon fell at his feet]

This example shows that the model is trained on data with a different lexicon (knees/feet, Simon/Peter) than that encountered in the example cases.

In particular, we can see in this example that the normalized text does not contain any abbreviations, but the name of Jesus is abbreviated by the

¹ All provided translations are our own, and we try to follow as closely the original text, sometimes at the expense of proper English wording.

Table 1: Summary of the evaluation data. The last column shows the percentage of vocabulary in the test dataset that also appears in the training dataset, and thus that has been seen by the model.

Dataset	Training data	Words (train)	Words (test)	% of train in test
CodGreek	Eclectic edition (SBLGNT)	19446	77701	98.87
CodLatin	Standardized translation (Vulgate)	14119	69021	91.76
Seals	Catalog transcription (Cheynet and Tatis (2019))	2843	2408	3.51

letters ω in the two considered manuscripts. This phenomenon is ubiquitous in biblical manuscripts, as scribes did not want to write out in full what was considered a "sacred name" (*Nomen Sacrum*) with religious significance (Metzger (1981)). Depending on the manuscript in question, the abbreviations used by scribes vary (Bezae, for example, uses three letters to abbreviate "Jesus," $\text{I}\eta\varsigma$). The impact of these abbreviated names on the performance of the segmentation is discussed in Section 9.

Despite these variations, as evidenced by the nearly 99% overlap between training and testing in Table 1, the Greek text of the New Testament has a relatively stable vocabulary across its textual witnesses, and the training data is fairly close to the test data, making this dataset the least challenging one.

All of our transcriptions were obtained from those provided by Institute for New Testament Textual Research (2024), and we use the SBLGNT text for training (Tauber (2017)).

Training on a standardized text (CodLatin)

The second use case also concerns a text from the New Testament, the Gospel of John, but this time in its Latin version. We consider seven codices, copied between the third and sixth centuries. This provides a test set of 4,560 Latin verses to be segmented in our evaluation dataset. These manuscripts present versions of the text known as the "Old Latin" (Vetus Latina), i.e., a collection of texts that preserve the great diversity of Latin Bible translations in circulation between the 2nd and 5th centuries AD. The Latin text was then standardized by the translator Jerome of Stridon, resulting in a version that would later become the official text of the Roman Catholic Church, known as the Vulgate (Houghton (2016)), and is now widely available in digital form. The relationship between the Old Latin texts and the Vulgate is still debated (Burton (2000)), and numerous morphological, syntactic, grammatical, and lexical

differences exist between them. For example, considering chapter 2, verse 2 of the Gospel of John in the Vulgate text and two of the codices chosen for our study:

- **Printed text: Vulgate:**

uocatus est autem ibi et iesus et discipuli eius ad nuptias [Then was called there and Jesus and his (eius) disciples to the wedding]

- **Codex 1: Palatinus:**

fuit autem ihs uocitus et discipuli eius ad nuptias [Then Jesus and his (eius) disciples had been called, to the wedding.]

- **Codex 2: Veronensis:**

inuitatus est autem et ihs cum discipulis suis ad nuptias [Then, Jesus was invited, with his (suis) disciples, to the wedding.]

This example shows how the training text (Vulgate) differs in terms of word order, vocabulary (including synonyms), grammar, and syntactic structure from the test texts found in the actual manuscripts. We also see, once again, the use of abbreviations for "sacred names", the word *iesus* being abbreviated by *ihs*. The linguistic divergence between the standardized training data (Vulgate) and the diverse test data (Vetus Latina) is therefore greater than in the previous case of the critical edition: the autoregressive model has to deal with a different lexicon, word order, and grammar than the data seen during training, as evidenced by the 91.76% overlap in vocabulary between the training and test datasets. All our transcriptions come from the electronic edition of the Vetus Latina for the Gospel of John (Burton et al. (2015)).

Training on catalog transcripts (Seals)

Finally, our last use case involves Byzantine seals, artifacts produced between the 6th and 12th centuries AD. We consider 353 seals from the Byzantine Empire, totaling 669 inscriptions in the test dataset, with most seals being double-sided. The Byzantine seals used in this study come from the

private collection of Mr. Yavuz Tatiş, a prominent Turkish businessman who assembled one of the most significant private collections of Byzantine seals in Turkey.

For training, we use the transcriptions from the corresponding published catalog (Cheynet and Tatiş (2019)). These catalog entries are full reconstructions of the seal texts, where the scholars have expanded all abbreviations and filled in missing or damaged parts. This significant difference between the training dataset and the test dataset makes this dataset the most difficult to segment in our study. For example, we can consider the difference between the transcription of seal 1752 (reverse side) from the Tatiş collection and its corresponding entry in the catalog edited by Cheynet and Tatiş (2019):

- **Printed text** (Seal 1752B):
Κύριε βοήθει τῷ σῷ δούλῳ Μιχαήλ ὀστιάριῳ
[Lord, save your servant Mickael Ostiarios]
- **Real artifact** (Seal 1752B):
Κε βῖ σῷ δού Μιχαήλ ὀστιαρ [Lord, save your servant Mickael Ostiarios]

This example illustrates the difficulties of segmenting the content of seals compared to codices: abbreviations are even more prevalent (Κε/Κύριε, δού/δούλῳ, ὀστιαρ(ίῳ)) as well as complete omissions of certain words such as definite articles (τῷ). This widespread use of abbreviations results in a low overlap between the test data and training data of 3.51%. The distance between the test set and the training set makes this dataset, by far, the most difficult to segment in our study.

5.2 Dataset annotation

We consider two possibilities for annotating the dataset used to train the models:

- **Binary classification scheme (I/E)**: if a character is at the end of the word, it is labeled as *E* (end of word), otherwise as *I* (inside).
- **Four-Class Classification (BIES)**: this annotation scheme, as proposed by Zhao et al. (2006) with the claim that it improves performance, considers that a character can be classified as B (beginning of a multi-character word), I (inside a multi-character word), E (end of a multi-character word), or S (single character).

We test these two annotation possibilities as hyperparameters of the model and discuss their impact on the system’s performance in Section 7. An example of annotation according to the two schemes considered is presented in Table 2.

6 Evaluation

6.1 Compared approaches

We evaluate our model against two state-of-the-art approaches, representing the two main paradigms for word segmentation.

1. **Bayesian Models (NHPYLM)** were the dominant paradigm for word segmentation before the advancements of neural models. We select the NHPYLM model due to its robustness, high quality of implementation, and widespread use.
2. **Supervised character-level sequence labeling approaches**, as used by Shao et al. (2018)) and applied to Latin in Clérice (2020). This approach corresponds to the first stage of our proposed method, and we use the same character-level model for sequence labeling. The purpose is two-fold: it will enable us to use it as the baseline for current approaches using single classification transformers, as well as serve as the basis for the ablation study presented in Section 7.4. We designate this model in the results section using the short name *Single-level*.

This selection was made to enable clear comparisons in this study between a canonical unsupervised method (NHPYLM), a flat neural model, and our proposed hierarchical neural method. This frames our benchmarking as a precise evaluation of the contribution of neural approaches compared to probabilistic ones, as well as demonstrating the usefulness of our novel hierarchical re-ranking stage compared to flat neural approaches.

6.2 Selected models and training procedure

Character-Level model

We implement character-level segmentation using the CANINE architecture (Clark et al. (2022)), a transformer specifically designed for direct

Table 2: Example of annotation schemes for word segmentation, taken from the first verse of the Gospel of John in the Vulgate.

Scriptio Continua	i	n	p	r	i	n	c	i	p	i	o	e	r	a	t	u	e	r	b	u	m
Ground truth	in		principio									erat				uerbum					
Schema 1 (Binary)	I	E	I	I	I	I	I	I	I	I	E	I	I	I	E	I	I	I	I	I	E
Schema 2 (4-class)	B	E	B	I	I	I	I	I	I	I	E	B	I	I	E	B	I	I	I	I	E

character-level text processing, thereby eliminating the need for subword tokenization. For each dataset, we fine-tuned a separate CANINE model from scratch, ensuring, for example, the model trained on the Greek text of Luke was not trained on the Byzantine seal data. The training data was randomly split into 80% for training and 20% for validation using a fixed random seed for reproducibility. Models were trained with the AdamW optimizer, with a maximum of 10 epochs or 10,000 steps. We employed early stopping based on the validation cross-entropy loss with a patience of 3 evaluation cycles. This procedure was repeated for each annotation scheme, resulting in one model per dataset and annotation type.

Word-Level model

For the re-ranking step, we rely on the XGLM model (Lin et al. (2022)), a large-scale, multilingual auto-regressive language model. The XGLM model was selected for the re-ranking stage due to its robust multilingual capabilities and proven performance in few-shot settings, which we considered important for handling the varied linguistic forms of our dataset.

We fine-tuned XGLM on the same training corpus used for the character-level model, with a causal language modeling objective. We trained the model for three epochs (1500 steps) with a learning rate of 1e-4, a sequence length of 128 tokens, and a stride of 64. Since XGLM relies on BPE for segmentation, unseen words do not present any particular problem for the computation of the perplexity.

The Word-level model is also trained from scratch on each dataset, producing three different models.

NHPYLM

For the NHPYLM algorithm (Mochihashi et al. (2009)), we used the implementation provided by Vojtěch Lanz (2025). Because the method is fully unsupervised and does not require ground

truth data, we train it directly on the evaluation data itself, rather than on the printed editions, as is the case for the transformer models. This training directly on the full dataset implies a natural advantage for this method that will learn the character and word distribution directly from the evaluation data. We adopted the default parameters from the library (Vojtěch Lanz (2025)).

6.3 Hyperparameters

For the analysis of the impact of hyperparameters, we investigate the effect of the beam width ($k \in \{5, 10, 15\}$) and the annotation labels (binary vs. 4-class). We perform a full grid search over these parameters, resulting in a total of six experiments considered for hyperparameter analysis. We do not investigate the impact of the architecture of each model, and leave this task for our further research work.

6.4 Evaluation scores

We draw from Goldwater et al. (2009) and Okabe et al. (2022) two categories of evaluation metrics: boundary-level metrics and word-level metrics.

Boundary level

To quantitatively evaluate the performance of word segmentation algorithms, we employ boundary-based metrics that assess the correspondence of the returned boundaries $\mathcal{B}_{pred} = \{\hat{b}_1, \hat{b}_2, \dots, \hat{b}_n\}$ with the actual boundaries $\mathcal{B}_{exp} = \{b_1, b_2, \dots, b_m\}$.

Given these two boundaries, we can define:

- **True Positives (TP):** Boundaries correctly identified by the algorithm, *i.e.* $TP = |\mathcal{B}_{exp} \cap \mathcal{B}_{pred}|$
- **False Positives (FP):** Boundaries that have been incorrectly predicted to measure over-segmentation, *i.e.* $FP = |\mathcal{B}_{pred} \setminus \mathcal{B}_{exp}|$

- **False Negatives (FN):** Boundaries that have been missed, to measure under-segmentation, *i.e.* $\text{FN} = |\mathcal{B}_{\text{exp}} \setminus \mathcal{B}_{\text{pred}}|$

We then rely on the following metrics to compute the quality of a given segmentation:

- **Precision:** Proportion of predicted boundaries that are correct:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- **Recall:** Proportion of true boundaries that are detected:

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- **F₁ score:** Harmonic mean of precision and recall:

$$F_1 = \frac{2PR}{P + R} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}.$$

Word level

At the word level, we evaluate the correctness of the extracted word tokens. We consider the precision (**word_precision**), which measures the proportion of predicted words that are correct, and recall (**word_recall**), which measures the proportion of correctly predicted words among the expected words. More formally, if we consider the ground truth segmentation $W_{\text{ref}} = \{w_1, w_2, \dots, w_k\}$ and the considered segmentation $W_{\text{pred}} = \{\hat{w}_1, \hat{w}_2, \dots, \hat{w}_L\}$, we define word-level precision and recall as:

$$\begin{aligned} \text{word_precision} &= \frac{\# \text{ of correctly segmented words}}{|W_{\text{pred}}|} \\ \text{word_recall} &= \frac{\# \text{ of correctly segmented words}}{|W_{\text{ref}}|} \end{aligned}$$

where a word is considered correct only if it matches the reference exactly in both form and boundaries.

6.5 Implementation and data availability

The character-level and word-level models were trained on a single consumer-grade machine, an Apple M3 Max (14 CPU cores, 30 GPU cores), with 36 GB of RAM. Training time for the character-level models lasted from 5 to 6 hours

per dataset, and from 10 to 14 hours for each of the word-level language models, depending on the dataset size.

To promote reproducibility and encourage widespread adoption of our solution, we have made all the code for this study available as an MIT open-source library on GitHub². This library includes a unified API for easily applying the three compared segmenters (our method, character-level labeling, and NHPYLM) to new texts.

All the experiments presented in this study can be reproduced using a dedicated **benchmark** command, ensuring full transparency of our study.

The data for the Codices in Greek and Latin are included in the companion repository. The data from the transcription seals, derived from the catalog by Cheynet and Tatiş (2019), is under copyright and is not distributed on our GitHub.

7 Results

7.1 Global results

The performance of our proposed solution on various benchmarking datasets, as well as for each combination of tested hyperparameters, is presented in Table 3.

As anticipated, performance varies depending on the difficulty of the dataset, due to the distribution shift between the training set (printed editions) and the test dataset (the transcriptions, or, in a full pipeline case, the output of the OCR). For the Greek and Latin datasets, boundary-level F1, precision, and recall exceed 93%, demonstrating high accuracy in boundary detection and, consequently, for post-OCR data cleaning. At the word level, the model also offers robust performance, achieving a precision of 92% and a recall of 91% for the **CodGreek** dataset, as well as a recall and precision of 89% for the **CodLatin** dataset. The balanced relationship between precision and recall across these datasets, especially in the case of **CodGreek**, indicates that the model does not have a bias towards over- or under-segmentation.

In contrast, the **Seals** dataset presents a greater challenge, as expected given the considerable distance between the training and test datasets: our

²<https://github.com/SphRbtHyk/ScriptioContinuaSegmenter>.

Table 3: Impact of hyperparameters on performance. Best performing hyperparameters in terms of F1 are highlighted in red.

Dataset	Annotation	Beam	F1 Score	Precision	Recall	Word precision	Word recall
CodGreek	Binary	3	0.95	0.96	0.95	0.90	0.89
		10	0.96	0.97	0.96	0.92	0.91
		15	0.96	0.97	0.96	0.92	0.91
	4-class	3	0.95	0.96	0.94	0.89	0.88
		10	0.96	0.97	0.95	0.91	0.89
		15	0.96	0.97	0.95	0.91	0.89
CodLatin	Binary	3	0.93	0.93	0.93	0.86	0.86
		10	0.94	0.94	0.94	0.88	0.88
		15	0.94	0.94	0.94	0.89	0.89
	4-class	3	0.92	0.93	0.92	0.85	0.85
		10	0.93	0.94	0.93	0.87	0.87
		15	0.93	0.94	0.93	0.88	0.88
Seals	Binary	3	0.30	0.44	0.23	0.44	0.38
		10	0.42	0.56	0.34	0.50	0.44
		15	0.42	0.53	0.45	0.48	0.42
	4-class	3	0.28	0.41	0.21	0.41	0.36
		10	0.34	0.49	0.26	0.46	0.40
		15	0.38	0.52	0.30	0.49	0.42

model achieves an F1-score of 42% for the best hyperparameter configuration. Word-level metrics indicate that 50% of the words segmented by the model are among the expected words (precision), with only 44% of the expected words detected (recall). This higher precision indicates that the model employs a conservative strategy, segmenting only words with recognizable forms, and thus misses some more ambiguous cases. While this represents a significant improvement over the current state of the art for the segmentation of the **Seal** dataset, as will be detailed in Section 7.3, these results indicate that further research is required before such algorithms are ready for fully automated use in sigillography.

7.2 Impact of hyperparameters

Regarding the impact of hyperparameters, we observe that the annotation scheme has a minimal impact on performance, and therefore does not confirm the claims found in Shao et al. (2018): 4-class annotation yields lower or equal performance compared to binary annotation, and multi-class classification introduces a slight performance loss over all scores. We therefore recommend using binary annotation alone, which both maximizes

performance and facilitates data preparation.

In contrast, the beam width has a significant impact on performance. Increasing the beam-width from 3 to 10 leads to F_1 gain of 1.01% for the Greek codices, 1.1% for the Latin codices, and a notable increase of 40% for Byzantine seals. However, increasing the width from 10 to 15 provides only a marginal gain, with execution time naturally rising as the number of beams increases. As the gain is marginal enough that the time/performance trade-off is not worthwhile, we consider 10 beams as the optimum configuration in terms of beam width. We therefore consider binary annotation with a beam width of 10 to be the recommended configuration for our experimental setup.

7.3 Comparison to the state of the art

The comparison between our solution and the state-of-the-art algorithms is presented in Table 4. The results show that our solution outperforms both of the compared baseline approaches. This superior performance is consistent across the considered benchmarking metrics.

Our hierarchical model achieves a significant gain

Table 4: Comparison to the state of the art. Best performance per dataset is highlighted in red.

Dataset	Algorithm	Annotation	F1 Score	Precision	Recall	Word precision	Word recall
CodGreek	Hierarchical		0.96	0.97	0.96	0.92	0.91
	Single Level	Binary	0.94	0.96	0.93	0.87	0.85
		4-class	0.94	0.96	0.93	0.87	0.85
	NHYPLM		0.68	0.64	0.74	0.34	0.40
CodLatin	Hierarchical		0.94	0.94	0.94	0.88	0.88
	Single Level	Binary	0.91	0.92	0.91	0.83	0.83
		4-class	0.91	0.92	0.91	0.84	0.83
	NHYPLM		0.71	0.66	0.79	0.43	0.51
Seals	Hierarchical		0.42	0.56	0.34	0.50	0.44
	Single Level	Binary	0.23	0.37	0.19	0.40	0.35
		4-class	0.24	0.38	0.19	0.40	0.35
	NHYPLM		0.32	0.27	0.50	0.20	0.24

in boundary-level F_1 performance, with a 2.1% improvement over character-level sequence labeling and a 41.2% improvement over the Bayesian approach on Greek codices. Similar gains are observed for the Latin codices, with respective gains of (3.3% and 32.4%), and on Byzantine seals (42.9% and 31.3%). The advantage of using a double-stage transformer is therefore greater as the dataset becomes more challenging.

The comparison between neural approaches, particularly those with hierarchical structures, and probabilistic methods shows that neural approaches are significantly more effective. This is especially notable given that the NHPYLM algorithm had the advantage of being trained directly on the test data, whereas the neural models were trained on printed editions. With a boundary detection F_1 ranging from 32% to 68%, the NHPYLM generates a high rate of false segments, resulting in an output that would require substantial manual correction when used in practice.

This difference in performance is even more pronounced for metrics calculated at the word level, with the Bayesian model consistently performing the worst in terms of correctly identifying words. For the Greek codices, precision increases from 34% with NHPYLM to 92% with our approach, and for the Latin codices, from 43% to 89%. For the Byzantine seals, precision improves from 20% to 50%.

This superior performance can be attributed to the capacity of neural models, which, compared to probabilistic models, can construct contextualized

character representations that enable the effective representation of the relationships between characters. The addition of the final re-ranking stage in our hierarchical approach ensures the global lexical coherence of the final segmentation, as confirmed by the improved performance of our approach compared to the single-level one.

7.4 Ablation study

The ablation study, which examines the benefits of our re-ranking second stage of the segmentation pipeline, is conducted by comparing the results between single-level and hierarchical transformers in Table 4. The hierarchical approach outperforms the single-level sequence labeling model across all benchmarking metrics considered.

For the boundary detection scores, the hierarchical approach always outperforms the single-level approach for all datasets and scores considered. We thus observe an increase in performance of 2.12% (**CodGreek**), 3.29% (**CodLatin**), and 75% (**Seals**). Thus, the more difficult the dataset and the greater the distance between the training datasets and the test dataset, the more efficient the second stage becomes.

The advantage of word-level re-ranking is even more pronounced when evaluating word-level precision scores, with gains of 5.74% (**CodGreek**), 5.95% (**CodLatin**), and 25% (**Seals**). This confirms that our re-ranking approach, which extracts segmentations with the highest syntactic quality, improves the model’s ability to correctly detect words.

8 Qualitative error analysis

For a qualitative error analysis, we manually reviewed the segmentation outputs for the Greek and Latin codices and identified the most common segmentation errors, classifying them into three categories: **(1)** Over and under-segmentation of composed words; **(2)** Non-detection of the abbreviations of the "Sacred Names"; **(3)** Wrongful segmentation of out-of-vocabulary words not seen in the training data.

We illustrate and discuss each category in detail in the following sections, providing illustrative examples and suggesting research directions to overcome these limitations. The **Seals** dataset, given its unique challenges and the lower performance of all considered models, is discussed in a separate section.

8.1 Examples of errors in the Codices

Composed words

A common source of error stems from the morphological complexity of Greek and Latin, where word fusion can lead to segmentation ambiguity, even for a scholar well-trained in these languages. Indeed, a frequently observed error in our resulting segmentations stems from the fact that Greek and Latin allow the addition of verbal prefixes, which nuance the meaning of verbs: for instance, adding the prefix ἐκ to the verb βάλλω (to throw) forms the word ἐκβάλλω (to throw out). However, these prefixes also exist independently as prepositions, with meanings of their own: for example, the prefix ἐκ corresponds to the preposition "out of." When faced with this phenomenon, the model, for example, segments ἐκβάλλω as ἐκ βάλλω. Among all word detection errors, 24% of them were related to this preposition prefixes phenomenon. One possible research venue to mitigate this phenomenon is to integrate syntactic knowledge into the segmentation process, for example, by considering part-of-speech (POS) tagging in the final decision stage.

Segmentation of "Sacred Names"

An anticipated limitation of our self-supervised training on modern editions is the handling of abbreviations, which are expanded in printed versions but are ubiquitous in codices. Since the

models have never been trained on these abbreviated forms, they consistently fail to segment them correctly.

This type of error accounts for 12% of cases observed in **CodGreek** and 2% in **CodLatin**. The higher prevalence of mistakes of this type in Greek codices can be attributed to the systematic use of abbreviations in Greek codices compared to Latin ones. When encountering these abbreviations, the model often fuses them with the neighboring word, creating incorrect segments. For example, the ground truth $\theta\upsilon \epsilon\iota\varsigma$, with $\theta\upsilon$ an abbreviation of $\theta\epsilon\omicron\upsilon$ ("of God"), is segmented as $\theta\upsilon\epsilon\iota\varsigma$, also an existing word meaning "you sacrifice." In Latin, the abbreviation *sps* for *spiritus* was never segmented correctly and was consistently merged with the following word. As the set of abbreviations is finite and well-documented in paleographic handbooks, a possible solution to this issue is to integrate a post-correction step that detects and expands these abbreviations using a rule-based heuristic.

Segmentation of out-of-vocabulary words

One of the main challenges in our approach is the domain shift introduced between the training and test datasets, as the training data is extracted from printed editions, making it difficult to segment out-of-vocabulary (OOV) words from the manuscripts. We thus observe that for the **CodGreek** dataset, 11% of errors are related to the incorrect segmentation of OOV words, compared to 15% in the case of the Latin dataset. This difference can be explained by the greater lexical divergence between the manuscripts containing *Vetus Latina* texts and the standardized *Vulgate* training text, while the Greek New Testament tradition presents a more stable vocabulary. Striking examples in Latin include the spelling of proper names: the name John is spelled as *Ioannes* in the *Vulgate*, but as *Iohannes* in *Codex Corbeien-sis Secundus*, leading to systematic segmentation errors. Regarding these proper name errors, an additional semantic layer implementing *Named Entity Recognition* (NER) could mitigate this issue. Our self-supervised approach, therefore, leads to a trade-off between performance and the cost of manual annotation, and a middle-ground could be the annotation of a portion of the considered artifacts.

8.2 The difficulties of the seal dataset

The **Seal** dataset highlights the current limitations of existing approaches when applied to fragmentary and abbreviated epigraphic texts, such as those found on seals or coins.

The main difficulty encountered by the model is its inability to segment ubiquitous sequences containing abbreviations that have never been encountered in the printed texts. Instead of inserting boundaries, the model prefers to leave the sequences fused. For example, the sequence Κε βθ τω σω , which should be segmented as Κε βθ τω σω ("Lord save your ..."), is left as a non-segmented string. This weakness due to the shift between the training and the testing dataset is clearly confirmed by the higher score obtained by the non-supervised approach NHPYLM compared to the single-level character-only labeling model. However, the model is effective at correctly segmenting whole words present in the transcription. For example, the word Κυριε ("Lord") when written in full, is consistently well segmented.

The main challenge for the **Seals** dataset is the significant change in vocabulary between the training and test sets. With only 3.51% of the tested words seen during training, the word-level reclassification model lacks the necessary context to evaluate candidate segmentations effectively. The character-level model, trained on extended forms, fails to recognize abbreviated character sequences as potential word boundaries. This creates a cumulative error effect that current architectures are unable to handle.

This task thus remains largely unsolved by current segmentation approaches. Among the considered possibilities, our research perspectives will include the addition of an "abbreviation" class to our character sequence tagging, along with a rule-based heuristic to expand abbreviations, which are often highly codified.

9 Limits

The qualitative error analysis has enabled us to detect four main limits of our current study, which will be addressed in future work:

1. **Dependence on linguistically similar training data:** One limitation of our self-supervised approach is the reliance on the availability of a printed text of the considered artifacts for the segmentation. For unedited texts or texts with significant linguistic differences, as clearly shown by the Byzantine seal dataset, where vocabulary overlap was only 3.51%, performance deteriorates significantly. Testing the transfer learning capacity of our approach is among our next working steps.
2. **Unexplored neural architectural choices:** Our study was limited to a single model architecture (CANINE and XGLM). Although these choices are justified and have yielded good performance, we did not perform an exhaustive comparison with other potentially suitable architectures, such as ByT5 (Xue et al. (2022)) for character tagging, or other large language models for the re-ranking stage. A study of the impact of the architecture on the performance of the segmentation task is part of our short-term work.
3. **Integration within the full-OCR pipeline:** One of the limitations of our study is the assumption of a character-accurate OCR input, as posited in *Assumption 1*. This problem framing was a deliberate choice to isolate and benchmark the already challenging word segmentation task and to check its feasibility on a perfect dataset, before addressing the complexity of the noise generated by OCR. Further work will involve testing the end-to-end pipeline and evaluating the impact of common OCR errors, such as character substitutions, insertions, or deletions, on the quality of the segmentation.
4. **Difficulty of working with Byzantine seals:** As discussed in Section 8.2, while our solution achieves excellent scores on the considered Greek and Latin codices, the scores are much lower on the Byzantine seal dataset. Therefore, although the performance on the Byzantine seals is a significant improvement over the state of the art, it still does not meet the required quality threshold for use in production by sigillography experts. Our study has thus demonstrated the limitations of current NLP methods when applied to highly abbreviated and fragmentary epigraphic texts, and the

need for the development of new, specialized approaches to such artifacts.

10 Conclusion and perspectives

In conclusion, this study presents a novel self-supervised hierarchical method for segmenting *Scriptio Continua* Greek and Latin texts. Our two-step approach, which combines character-level sequence labeling and sequence re-ranking using an auto-regressive language model perplexity score, significantly improves the current baseline for word segmentation. The proposed method achieved an F_1 score for word boundary detection of 96% (4 Greek codices), 94% (7 Latin Codices), and 42% (699 inscriptions on Byzantine Seals), and a precision of 92%, 89%, and 50%, respectively, for the proper detection of words. This novel approach systematically outperforms the current baselines found in the state of the art, leading to F_1 gains of 2.1% for word boundary detection compared to standard labeling models and of 41.2% for NHPYLM on Greek codices, 3.3% and 32.4% on Latin codices, and 42.9% and 31.3% on Byzantine seals. In terms of precision for proper word detection, our method increases precision from 34% with NHPYLM to 92% for the Greek codices, and from 43% to 89% for the Latin codices. For the Byzantine seals, precision improves from 20% to 50%. Consequently, our approach proves to be a promising tool for the segmentation step in the post-OCR processing models' output applied to texts written in *Scriptio Continua*, provided that high-quality character recognition is achieved. This study also enabled us to demonstrate the need for the development of specific methods for studying abbreviated and fragmentary epigraphic texts, and that the segmentation of Byzantine seals remains an open challenge.

Besides working on the potential further research directions presented in Section 9, our next step is to integrate our segmentation model into a complete pipeline that can process artifact images directly into post-processed, segmented text.

Declarations

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